Answers learning task 1

2026 – 2027

In Questions 2.1, 2.8 and 2.17, provide a reasonable description of the sample space for each of the random experiments. There can be more than one acceptable interpretation of each experiment. Describe any assumptions you make.

Exercise 2.1: Each of three machined parts is classified as either above or below the target specification for the part.

Answer

The sample space can be described as

$$S = \{aaa, aab, aba, baa, abb, bab, bba, bbb\},$$

where $a =$ above and $b =$ below.

Exercise 2.8: The following two questions appear on an employee survey questionnaire. Each answer is chosen from the five-point scale 1 (never), 2, 3, 4, 5 (always).

- Is the corporation willing to listen to and fairly evaluate new ideas?
- How often are my coworkers important in my overall job performance?

Answer

The sample space can be described as

$$S = \{(1,1), (1,2), \ldots, (1,5), (2,1), \ldots, (5,5)\}$$

The number of possible outcomes in S equals $n_S = 5 \cdot 5 = 25$.

Exercise 2.17: Calls are repeatedly placed to a busy phone line until a connection is achieved.

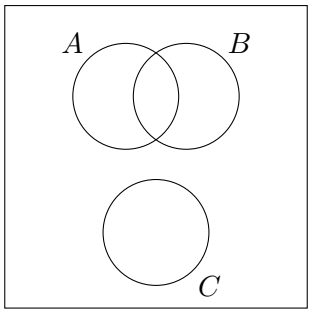
Answer

The sample space can be described as

$$S = \{c, bc, bbc, bbbc, \ldots\},$$

where $b =$ busy and $c =$ connect.

Exercise 2.20: Three events are shown on the Venn diagram in the following figure:

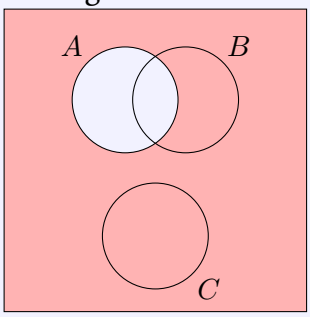


Reproduce the figure and shade the region that corresponds to each of the following events.

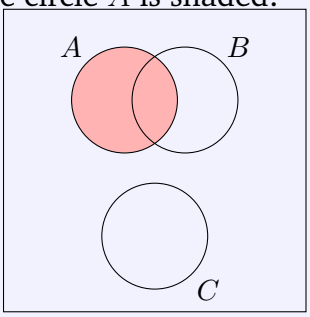
(a) A'
(b) $(A \cap B) \cup (A \cap B')$
(c) $(A \cap B) \cup C$
(d) $A \cap (B \cup C)'$
(e) $(A \cap B)' \cup C$

Answer

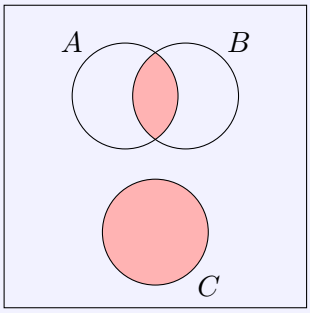
(a) A' is the region outside circle A:



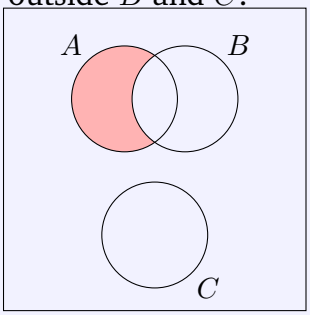
(b) $(A \cap B) \cup (A \cap B')$ simplifies to A, so the entire circle A is shaded:



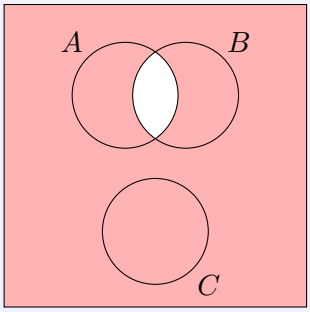
(c) $(A \cap B) \cup C$ is the intersection of A and B, plus the entire circle C:



(d) $A \cap (B \cup C)'$ is the portion of circle A that is outside B and C:



(e) $(A \cap B)' \cup C$ is everything outside the intersection of A and B, plus the entire circle C:



Exercise 2.29: The rise time of a reactor is measured in minutes (and fractions of minutes). Let the sample space be positive, real numbers. Define the events A and B as follows: $A =$

$\{x \mid x < 65\}$ and $B = \{x \mid x > 45.5\}$.

Describe each of the following events:

(a) A' (b) B'
(c) $A \cap B$ (d) $A \cup B$

Answer

(a) $\{x \mid x \geq 65\}$ (b) $\{x \mid 0 < x \leq 45.5\}$
(c) $\{x \mid 45.5 < x < 65\}$ (d) $\{x \mid 0 < x < \infty\}$

Exercise 2.31: A sample of two printed circuit boards is selected without replacement from a batch. Describe the (ordered) sample space for each of the following batches:

(a) The batch contains 90 boards that are not defective, seven boards with minor defects, and three boards with major defects.

(b) The batch contains 90 boards that are not defective, eight boards with minor defects, and one board with major defects.

Answer

a. $S = \{gg, gm, gb, mg, mm, mb, bg, bm, bb\}$,
where $g =$ good, $m =$ minor and $b =$ bad
b. $S = \{gg, gm, gb, mg, mm, mb, bg, bm\}$

Exercise 2.55: The sample space of a random experiment is $\{a, b, c, d, e\}$ with probabilities $0.2, 0.1, 0.2, 0.3,$ and 0.2, respectively. Let A denote the event $\{a, b, c\}$, and let B denote the event $\{c, d, e\}$. Determine the following:

(a) $P(A)$ (b) $P(B)$ (c) $P(A')$
(d) $P(A \cup B)$ (e) $P(A \cap B)$

Answer

1. $P(A) = P(a) + P(b) + P(c) = 0.2 + 0.1 + 0.2 = 0.5$
2. $P(B) = P(c) + P(d) + P(e) = 0.2 + 0.3 + 0.2 = 0.7$
3. $P(A') = 1 - P(A) = 1 - 0.5 = 0.5$
4. Notice that $A \cup B = \{a, b, c, d, e\} = S$, therefore $P(A \cup B) = 1$.
5. Notice that $A \cap B = \{c\}$, therefore $P(A \cap B) = P(c) = 0.2$.

Exercise 2.74: If $P(A) = 0.4$, $P(B) = 0.2$, and $P(A \cap B) = 0.1$, determine the following probabilities:

(a) $P(A')$ (b) $P(A \cup B)$ (c) $P(A' \cap B)$
(d) $P(A \cap B')$ (e) $P[(A \cup B)']$ (f) $P(A' \cup B)$

Answer

1. $P(A') = 1 - P(A) = 1 - 0.4 = 0.6$
2. $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.4 + 0.2 - 0.1 = 0.5$
3. $P(A' \cap B) = P(B) - P(A \cap B) = 0.2 - 0.1 = 0.1$
4. $P(A \cap B') = P(A) - P(A \cap B) = 0.4 - 0.1 = 0.3$
5. $P[(A \cup B)'] = 1 - P(A \cup B) = 1 - 0.5 = 0.5$
6. $P(A' \cup B) = P(A') + P(B) - P(A' \cap B) = 0.6 + 0.2 - 0.1 = 0.7.$

Exercise 2.77: Disks of polycarbonate plastic from a supplier are analyzed for scratch and shock resistance. The results from 100 disks are summarized as follows:

		Shock Resistance	
		High	Low
Scratch	High	70	10
Resistance	Low	15	5

(a) If a disk is selected at random, what is the probability that its scratch resistance is high and its shock resistance is high?

(b) If a disk is selected at random, what is the probability that its scratch resistance is high or its shock resistance is high?

(c) Consider the event that a disk has high scratch resistance and the event that a disk has high shock resistance. Are these two events mutually exclusive?

Answer

(a) Of the 100 disks, there are exactly 70 with high scratch resistance and high shock resistance (left top cell). Therefore the probability of picking such a disk is

$$\frac{\text{number of favorable outcomes}}{\text{total number of outcomes}} = \frac{70}{100} = 0.7.$$

(b) Of the 100 disks, there are exactly $70 + 10 + 15 = 95$ with high scratch resistance or high shock resistance (all cells in left column or top row). Therefore the probability of picking such a disk is

$$\frac{\text{number of favorable outcomes}}{\text{total number of outcomes}} = \frac{95}{100} = 0.95.$$

(c) Let A be the event of high scratch resistance, and B the event of high shock resistance. We have that $P(A \cap B) = 0.7 > 0$, hence A and B are not mutually exclusive.

Exercise 2.87: Disks of polycarbonate plastic from a supplier are analyzed for scratch and shock resistance. The results from 100 disks are summarized as follows:

		Shock Resistance	
		High	Low
Scratch Resistance	High	70	10
	Low	15	5

Let A denote the event that a disk has high shock resistance, and let B denote the event that a disk has high scratch resistance. Determine the following probabilities:

(a) $P(A)$
(b) $P(B)$
(c) $P(A \mid B)$
(d) $P(B \mid A)$

Answer

1. Of the 100 disks, there are exactly $70 + 15 = 85$ with high shock resistance (left column). Therefore:

$$P(A) = \frac{\text{number of favorable outcomes}}{\text{total number of outcomes}} = \frac{85}{100} = 0.85.$$

2. Of the 100 disks, there are exactly $70 + 10 = 80$ with high shock resistance (top row). Therefore:

$$P(B) = \frac{\text{number of favorable outcomes}}{\text{total number of outcomes}} = \frac{80}{100} = 0.80.$$

3. Of the 80 disks with high scratch resistance, there are exactly 70 with high shock resistance (top left cell). Therefore:

$$P(A \mid B) = \frac{\text{number of favorable outcomes}}{\text{total number of outcomes}} = \frac{70}{80} = 0.875.$$

4. Of the 85 disks with high shock resistance, there are exactly 70 with high scratch resistance (top left cell). Therefore:

$$P(B \mid A) = \frac{\text{number of favorable outcomes}}{\text{total number of outcomes}} = \frac{70}{85} = 0.8235.$$

Exercise 2.91: The following table summarizes the analysis of samples of galvanized steel for coating weight and surface roughness:

		Coating Weight	
		High	Low
Surface Roughness	High	36	16
	Low	114	34

(a) If the coating weight of a sample is high, what is the probability that the surface roughness is high?

(b) If the surface roughness of a sample is high, what is the probability that the coating weight is high?

(c) If the surface roughness of a sample is low, what is the probability that the coating weight is low?

Answer

Let us define the events

$$A = \{\text{Coating weight is high}\}, B = \{\text{Surface roughness is high}\}.$$

The requested probabilities are as follows:

(a) $P(B \mid A) = \frac{P(B \cap A)}{P(A)} = \frac{36}{36+114} = 0.24.$

(b) $P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{36}{36+16} = 0.6923.$

(c) $P(A' \mid B') = \frac{P(A' \cap B)}{P(B')} = \frac{34}{34+114} = 0.2297.$

Exercise 2.106: Suppose that $P(A \mid B) = 0.25$, $P(A \mid B') = 0.35$, and $P(B) = 0.8$. What is $P(A)$?

Answer

By the complement rule, we have that $P(B') = 1 - P(B) = 1 - 0.8 = 0.2$. Using the law of total probability, we get:

$$P(A) = P(A \mid B) \cdot P(B) + P(A \mid B') \cdot P(B') = 0.25 \cdot 0.8 + 0.35 \cdot 0.2 = 0.27.$$

Exercise 2.113: A batch of 30 injection-molded parts contains 6 parts that have suffered excessive shrinkage.

(a) If two parts are selected at random, and without replacement, what is the probability that the second part selected is one with excessive shrinkage?

(b) If three parts are selected at random, and without replacement, what is the probability that the third part selected is one with excessive shrinkage?

Answer

Let us define for $i = 1, 2, 3$ the events $S_i = \{\text{the } i\text{-th part has excessive shrinkage}\}$. We know that $P(S_1) = \frac{6}{30} = \frac{1}{5}$ and can find the probabilities $P(S_2)$ and $P(S_3)$ as follows:

(a) To compute $P(S_2)$, we can split it in two cases (based on whether or not S_1 occurred):

- If S_1 occurred, then for the second pick we have $30 - 1 = 29$ options of which $6 - 1 = 5$ have excessive shrinkage. With other words: $P(S_2|S_1) = \frac{5}{29}$.
- If S_1 did not occur, then for the second pick we have $30 - 1 = 29$ options of which $6 - 0 = 6$ have excessive shrinkage. With other words: $P(S_2|S_1') = \frac{6}{29}$.

Using the law of total probability, we get:

$$P(S_2) = P(S_2 \mid S_1) \cdot P(S_1) + P(S_2 \mid S_1') \cdot P(S_1') = \frac{5}{29} \cdot \frac{1}{5} + \frac{6}{29} \cdot \frac{4}{5} = \frac{29}{145} = 0.2.$$

(b) To compute $P(S_3)$, we can split it in four cases (based on whether or not S_1 and/or S_2 occurred):

- If S_1 and S_2 occurred, then for the third pick we have $30 - 2 = 28$ options of

which $6-2=4$ have excessive shrinkage. With other words: $P(S_3|S_1 \cap S_2) = \frac{4}{28}$.

- If S_1 did not occur, but S_2 did, then for the third pick we have $30-2=28$ options of which $6-1=5$ have excessive shrinkage. With other words: $P(S_3|S_1' \cap S_2) = \frac{5}{28}$.
- If S_1 occurred, but S_2 did not, then for the third pick we have $30-2=28$ options of which $6-1=5$ have excessive shrinkage. With other words: $P(S_3|S_1 \cap S_2') = \frac{5}{28}$.
- If neither S_1 nor S_2 occurred, then for the third pick we have $30-2=28$ options of which $6-0=6$ have excessive shrinkage. With other words: $P(S_3|S_1' \cap S_2') = \frac{6}{28}$.

Using the law of total probability, we get:

$$\begin{aligned} P(S_3) &= P(S_3 \mid S_1 \cap S_2) \cdot P(S_1 \cap S_2) + P(S_3 \mid S_1' \cap S_2) \cdot P(S_1' \cap S_2) \\ &\quad + P(S_3 \mid S_1 \cap S_2') \cdot P(S_1 \cap S_2') + P(S_3 \mid S_1' \cap S_2') \cdot P(S_1' \cap S_2') \\ &= \left(\frac{4}{28} \cdot \frac{6}{30} \cdot \frac{5}{29}\right) + \left(\frac{5}{28} \cdot \frac{24}{30} \cdot \frac{6}{29}\right) + \left(\frac{5}{28} \cdot \frac{6}{30} \cdot \frac{24}{29}\right) + \left(\frac{6}{28} \cdot \frac{24}{30} \cdot \frac{23}{29}\right) \\ &= 0.2. \end{aligned}$$